For this reason, we will be doing the same thing here and use the built-in UCB policy class, even though we could simply compute the quantity $\mu + \beta\sigma$ ourselves. This can be done with:

```
policy = botorch.acquisition.analytic.UpperConfidenceBound(
    model, beta=1
)
```

Here, the UCB class implementation in BoTorch takes in the GP model as its first input and a positive value for its second input `beta`. As you might expect, this second input denotes the value for the UCB parameter $\beta$ in the score formula $\mu + \beta\sigma$, which trades off between exploration and exploitation. Here, we will set it to 1 for now.

**NOTE** **Believe it or not, this is all we need to change from our BayesOpt code from the previous chapter, in order to run the UCB policy on the objective functions we have. This demonstrates the benefit of BoTorch's modularity, which allows us to plug and play any policy we'd like to use into our BayesOpt pipeline.**

Running our UCB policy with $\beta = 1$ on our familiar Forrester objective function generates Figure 5.9, which shows optimization progress throughout ten function evaluations.

We see that similar to what happened to the Probability of Improvement policy, UCB with $\beta = 1$ fails to sufficiently explore the search space and is stuck at a local optimum. This means that the value for $\beta$ is too small.

**SIDEBAR** **SIDEBAR**

**We learn about the Probability of Improvement policy in Section 4.2.**

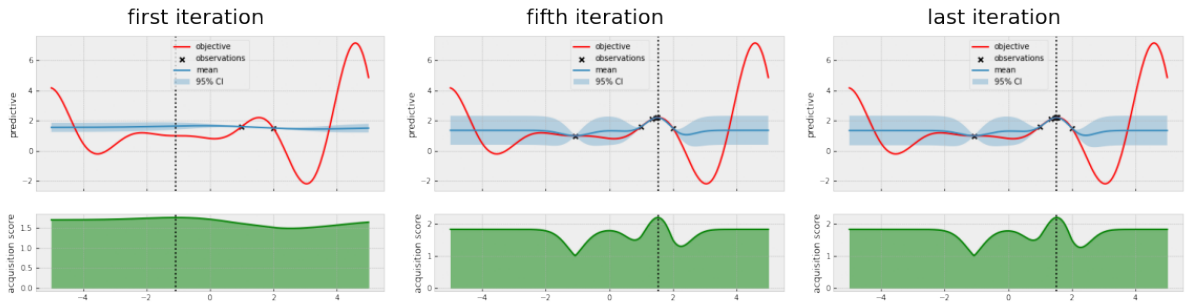



**Figure 5.9 Progress made by the Upper Confidence Bound policy with the tradeoff parameter = 1. The value of the parameter is not sufficiently large to encourage exploration, causing progress to be stuck at a local optimum.**

Let us try again, this time setting this tradeoff parameter to a larger value:

```
policy = botorch.acquisition.analytic.UpperConfidenceBound(
    model, beta=2
)
```

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